Google Ads Autopilot: MCP and Agent PRD

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Summary

Build a private, single-user MCP server that lets Claude monitor a Google Ads account around the clock and change it only inside limits the server enforces. Nothing depends on Claude being available to stop overspend: a deterministic watchdog runs without it.

A single Claude session cannot stay open indefinitely, so "constant" comes from two parts. A watchdog checks spend and health every 15 minutes, and scheduled Claude runs do the judgment work hourly, daily and weekly.

Decision	Choice	Why
Hosting	Remote Streamable HTTP server on the owner's GKE or Cloud Run	Works from the Claude mobile app and in unattended runs
Stack	Java 21, Spring Boot, Spring AI, Google Ads Java client	Matches the owner's existing MCP stack
Launch scope	Search campaigns on one allowlisted account	Smallest surface where search-term and keyword tools add value
Who can apply changes	Claude applies low-risk changes within limits; everything else needs human approval	Limits the cost of a bad decision
Approval path	Signed link outside Claude, behind Google sign-in	Claude has no tool that can approve its own proposal
Stop mechanism	Kill switch and automatic circuit breaker in the server	Works even if a Claude run is wrong or compromised

Goals, non-goals and success metrics

The product succeeds if it improves results per unit of spend while making an expensive mistake hard to cause. Targets below are starting points; performance targets get set after a two-week read-only baseline.

Goals

- Detect spend, delivery and policy problems without the owner watching the account.
- Remove repetitive work such as search-term review and negative keywords.
- Let the owner create and adjust campaigns from a Claude conversation.
- Record every change with its reason, its before and after values, and a rollback path.

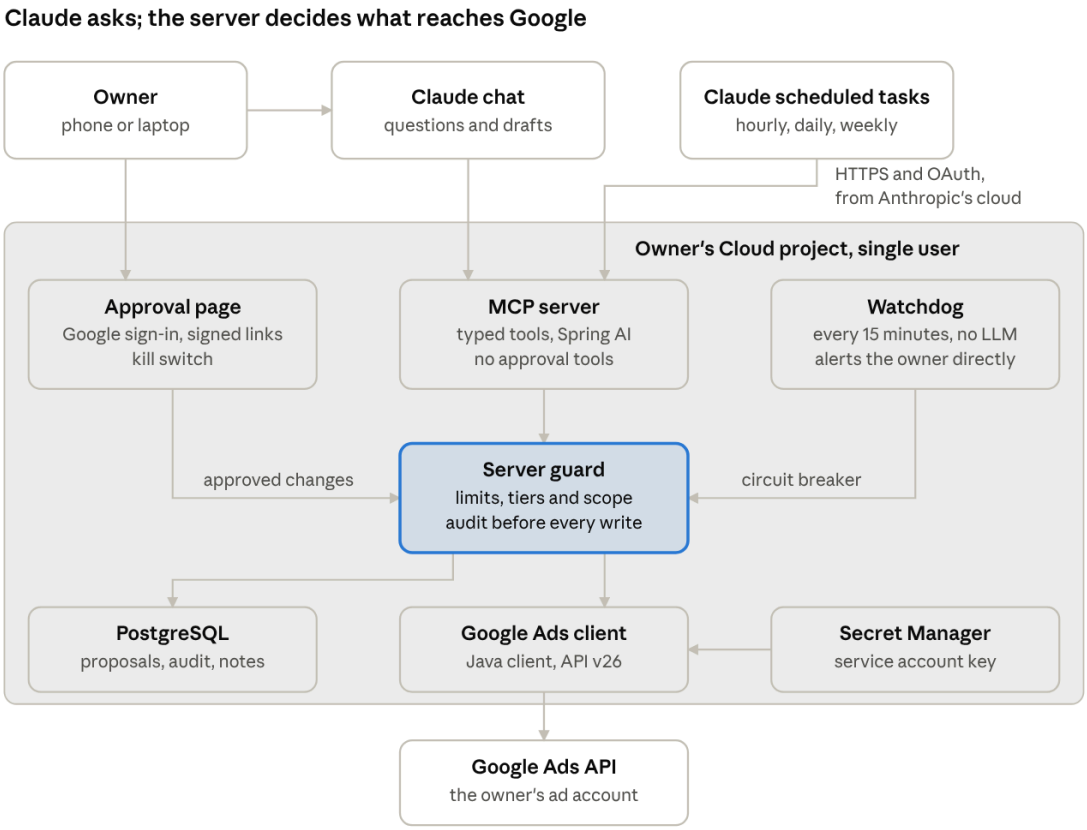
Non-goals for v1

- Hosting for other people or businesses. Google's policy makes this a separate compliance question.
- Billing, payment, user management or account creation changes.
- Offline conversion and Customer Match uploads, which are restricted for new API adopters.
- Performance Max, Shopping, Display, Video and other ad platforms.
- Browser automation of the Google Ads interface.

Metric	Target	Measured by
Changes outside hard limits	0	Guard rejections and audit log review
Changes with reason, diff and audit record	100%	Audit log
Time to alert on a spend breach	Under 30 minutes	Injected test breach in the test account
Watchdog uptime	99% of scheduled runs per month	Heartbeat monitor
Agent-applied changes rolled back by the owner	Under 10%	Audit log
Spend on irrelevant search terms	Down from baseline; target set after week 2	Weekly report
Cost per conversion	At or better than baseline; target set after week 2	Weekly report, compared with the same period before launch
Owner time on routine account work	Down from baseline	Owner's estimate in the monthly review

Architecture

All writes pass through one guard inside the owner's Cloud project, whichever part of the system asked for them.



system architecture · Claude, MCP server, guard, watchdog, approval page and Google Ads

Claude reaches the server only through MCP tools, and those tools cannot approve a proposal. Approvals come through a separate page that requires the owner's Google sign-in. The watchdog needs no Claude session, so alerts and the circuit breaker keep working if a scheduled run fails.

The server is one Spring Boot codebase with three entry points: MCP tools, the approval page and the watchdog job. Sharing one guard and one audit log means a safety rule only has to be built and tested once.

Agent operating model

Five loops split the work so that deterministic checks run often and Claude runs when judgment is needed. Each scheduled Claude run starts as a fresh session, so goals, memory and open items live on the server and load at the start of each run.

Loop	Cadence	Runs on	Job	Allowed to change
Watchdog	Every 15 minutes	Server, no LLM	Check spend against caps, disapprovals, stopped delivery, unexpected changes and quota	Only pause through the circuit breaker

Loop	Cadence	Runs on	Job	Allowed to change
Triage	Hourly	Claude scheduled task	Diagnose open alerts and decide what they need	Emergency pause when a cap is breached; otherwise proposals
Daily optimization	Daily, early morning Warsaw time	Claude scheduled task	Review search terms, keywords, ads and pacing; check earlier changes	Tier 1 changes at autonomy level 2 and above
Weekly review	Monday morning	Claude scheduled task	Report results, propose budget moves, ad tests and structure changes	Proposals only
On demand	When the owner asks	Claude chat, including mobile	Answer questions and draft campaigns and ads	Proposals only

Scheduled tasks normally run at most hourly, so anything faster belongs to the watchdog. If hourly runs use too much of the Claude plan's allowance, the fallback is a Claude Agent SDK worker that the watchdog wakes only when an alert opens.

Run contract

Every scheduled run follows the same steps, from a versioned prompt kept in the repository:

1. Call `get_agent_context` for goals, limits, autonomy level, open alerts, pending proposals and notes from earlier runs.
2. Read only the data the loop needs, through purpose-built tools.
3. Check whether earlier changes produced the expected effect.
4. Apply allowed Tier 1 changes or file proposals, each with a reason and expected effect.
5. Call `record_run_summary` with what it checked, what it did and what it wants the next run to look at.

Autonomy levels

The owner sets the level in server configuration. Claude can read it but has no tool to change it.

Level	Name	What Claude can do	Entry condition
L0	Observe	Read and report	Launch
L1	Propose	File proposals for the owner to approve	Read-only phase gate passed
L2	Limited autonomy	Apply Tier 1 changes within limits; everything else stays a proposal	Proposal phase gate passed

Level	Name	What Claude can do	Entry condition
L3	Extended autonomy	Tier 1 limits widened by the owner, one change type at a time	At least 8 weeks at L2 with the rollback target met

Creating ads, enabling campaigns and raising budgets stay at Tier 2 at every level.

Decision rules

- Recent conversion data can be incomplete, so performance decisions exclude the latest 3 days. Spend and delivery checks do not.
- Claude does not change an entity again during its cooldown, so each result can be read before the next change.
- Claude touches only entities labelled `agent-managed`. A `no-agent` label always wins.
- Changes made by the owner in the Google Ads interface are never reverted. Claude reads them as the new baseline and notes them in its summary.

MCP tool catalog

The server exposes narrow, purpose-built tools rather than a generic mutate call, so every write can be checked against a known rule. Risk tier is decided by the server per call, not by Claude.

Tier	Meaning	Who can trigger it
T0	Read or internal record; no account change	Claude
T1	Low-risk change inside hard limits	Claude applies at L2 and above; a proposal at L1
T2	Meaningful spend or brand effect	Claude proposes; only the owner approves
T3	Not exposed through MCP	Owner, in the Google Ads interface or server configuration

Read and context tools (T0)

Tool	Returns
`get_agent_context`	Goals, limits, autonomy level, open alerts, pending proposals, recent run notes
`get_account_health`	Spend against caps, pacing, delivery status, disapprovals, quota used
`get_performance`	Metrics by campaign, ad group, keyword or ad for a date range

Tool	Returns
`get_search_terms`	Search terms with cost, clicks, conversions and current negative coverage
`get_keywords`	Keywords with status, match type, bid and quality data
`get_ads`	Responsive search ads with assets, ad strength and policy status
`get_change_history`	Recent account changes by source, including changes outside this system
`get_recommendations`	Google recommendations, for reading only
`run_report`	A read-only GAQL query, limited to allowlisted resources and a row cap
`list_proposals` / `get_proposal`	Proposal status, diff and outcome
`record_run_summary`	Saves run notes for the next session

Change tools

Every change tool requires `reason`, `expected_effect`, `run_id` and `idempotency_key`.

Tool	Tier	Rule
`add_negative_keywords`	T1	Exact or phrase match; blocked if it matches the protected term list
`pause_keyword`	T1	Needs the minimum data threshold; ad group keeps at least one active keyword
`adjust_keyword_bid`	T1	Manual CPC only; within the per-change limit
`decrease_budget`	T1	Within the per-change limit
`emergency_pause_campaign`	T1	Only while a spend cap is breached
`pause_ad`	T2	Ad group keeps at least one enabled ad
`add_keywords`	T2	Match type and ad group shown in the diff
`create_ad`	T2	3 to 15 headlines, 2 to 4 descriptions, final URL on the allowed domain list
`create_campaign`	T2	Always created `PAUSED`, with budget, targeting and ad group in one atomic request
`increase_budget`	T2	Within the per-change limit and all spend caps

Tool	Tier	Rule
`change_bidding_strategy`	T2	Diff shows old and new strategy
`enable_entity`	T2	Starts delivery, so it is always approved separately
`apply_recommendation`	T2	One at a time, with the real change shown in the diff
`propose_rollback`	T2	Restores recorded earlier values

Deliberately absent (T3)

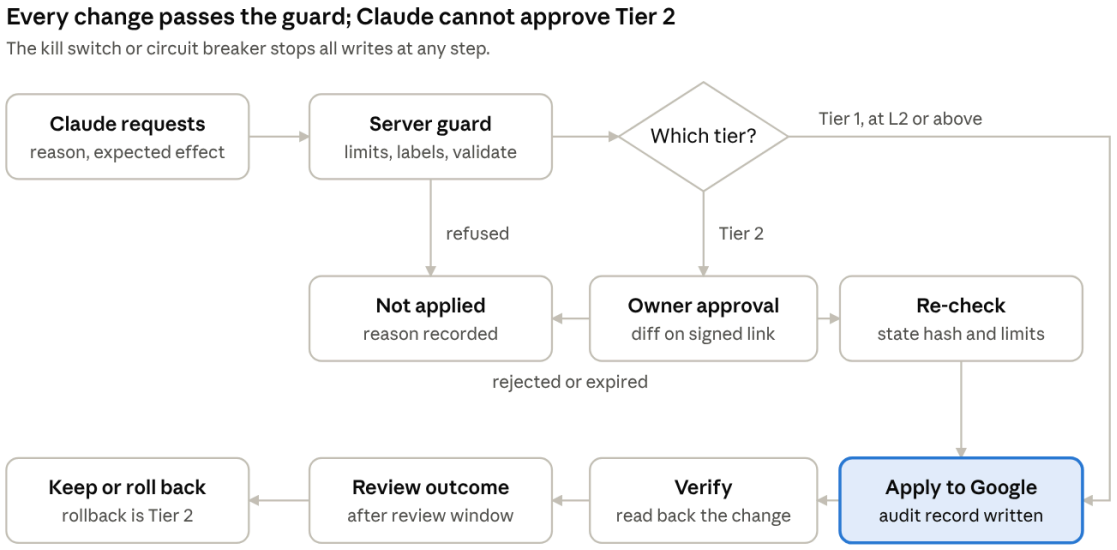
- **Approval and control:** approving proposals, changing limits or autonomy, and turning off the kill switch.
- **Generic writes:** any raw mutate tool, or a tool that accepts arbitrary operations.
- **Account administration:** billing, payment, users, account links and account creation.
- **Destructive or tracking changes:** removing campaigns, ad groups or ads, and changing conversion actions.

Tool design rules

- Input is typed and validated, with money in account currency micros and IDs checked against the allowlisted account.
- Results are compact, paged and sorted by cost, with a clear note when rows are cut off.
- Refusals name the rule and the limit, for example: "Budget change of +35% exceeds the +20% limit", so Claude can adjust or file a proposal.
- Repeating a call with the same `idempotency_key` returns the earlier result instead of changing the account twice.

Guardrails and change lifecycle

Limits live in the server, outside Claude's reach, so a wrong conclusion is refused instead of executed. All values below are proposed defaults for the owner to confirm, and writes stay disabled until the required caps are set.



change lifecycle · guard, tier decision, approval, apply, review

Tier 1 changes go straight from the guard to Google. Tier 2 changes wait for the owner, and the server re-checks them on approval because the account may have changed in the meantime.

Hard limits

Limit	Proposed default	Enforced on
Monthly account cap	Owner sets; required before any write	All budget changes and campaign creation
Sum of daily budgets × 30.4	Must stay at or under the monthly cap	All budget changes and campaign creation
Budget increase	Up to +20% per change, once per 7 days, always approved	`increase_budget`
Budget decrease	Up to −20% per change, once per 24 hours	`decrease_budget`
Bid change	Up to ±10% per change, once per 7 days per keyword	`adjust_keyword_bid`
Negative keywords	Up to 25 per day; exact or phrase match	`add_negative_keywords`
Keyword pause threshold	At least 100 clicks and 0 conversions in 30 days	`pause_keyword`
Tier 1 volume	Up to 50 changes per day	All Tier 1 tools

Limit	Proposed default	Enforced on
Cooldown	No second change to the same entity within 72 hours, except emergency pause	All change tools
Scope	Allowlisted account and `agent-managed` label only	All change tools
Destinations	Final URLs on the allowed domains; targeting within the allowed locations	Ads and campaigns
Proposal lifetime	Expires after 48 hours; invalid if the entity changed	All Tier 2 tools

Stop mechanisms

- **Daily alert:** The watchdog alerts when spend reaches 1.5 times the summed daily budgets. Google permits up to 2 times on a given day, so this is a warning and not automatically a fault.
- **Circuit breaker:** The watchdog pauses `agent-managed` campaigns and blocks writes when spend reaches 1.8 times the summed daily budgets or month-to-date spend reaches the monthly cap.
- **Kill switch:** A single setting makes the server read-only at once. It works from the approval page and from server configuration, not through Claude.
- **Resume:** Only the owner can clear a breaker or kill switch, and the campaigns the breaker paused are listed for review.
- **Failure mode:** If the database, limits or audit log are unavailable, writes are refused and reads continue.

Audit and rollback

Every attempt, including refused ones, stores the run, reason, old and new values, tier, approver and Google request ID. Rollback restores recorded values through a new proposal, so it follows the same checks and leaves its own audit trail. Rollback cannot restore lost delivery time, so the owner still reviews any pause that affects active campaigns.

Security and policy compliance

The main threat is a wrong or manipulated Claude run spending money, so the server treats every tool call as untrusted and limits what a compromised run can do.

Authentication

Link	Method	Requirement
Server to Google Ads	Service account in the owner's own Cloud project	Google recommends service accounts over a single-user token. Add it as a Standard user, not an Admin. Keep the key in Secret Manager.

Link	Method	Requirement
Claude to server	OAuth over HTTPS	Restrict sign-in to the owner's Google account. Support token expiry and refresh, because Claude handles both.
Owner to approval page	Google sign-in plus a signed, single-use link	Links expire with the proposal. The page shows the exact diff, and the server re-checks limits on approval.

Connector traffic comes from Anthropic's cloud, not the owner's device, so the endpoint must be reachable from the internet. Add rate limits and allowlist Anthropic's published IP ranges as a second layer.

A separate service account means API changes appear under their own identity in Google's change history, so they are easy to tell apart from the owner's.

Prompt injection

Search terms are typed by strangers and reach Claude as data, so they could carry instructions. Ad text, page content and recommendation text carry the same risk.

- Mark free-text fields in tool results as untrusted data, and keep them apart from system fields.
- Keep the server's limits independent of anything Claude read or concluded.
- Give the agent no tool that fetches arbitrary URLs or sends messages to arbitrary recipients.
- Include injection strings in the test fixtures, such as a search term telling the agent to raise a budget.
- Filter search terms that look like instructions into a review list, so they never become keywords automatically.

Secrets and data

- Never pass credentials through tool arguments, results, prompts or logs.
- Store aggregated campaign data only; no customer lists or personal data are in scope.
- Keep audit records for at least 24 months and metric snapshots for 13 months, so year-over-year comparisons work.

Google policy

The personal, self-run shape of this product is what keeps it on the allowed side of Google's 31 August 2026 policy. This reading comes from a secondary source, so the owner should confirm it against the policy definition before launch.

Policy point	How this design responds
Hosted third-party layers that only re-expose the API are banned, and MCP servers are named	Runs for the owner, on the owner's infrastructure, against the owner's account, with no other users

Policy point	How this design responds
A developer's own automation on their own accounts is exempt	Matches the v1 scope exactly
Each integration needs its own dedicated Cloud project	Uses one project created only for this server
Shared credentials and access passed to third parties are prohibited	No external users; credentials never leave the owner's project
Headless access for end users needs per-entity Google authentication	There are no end users; the owner's account is the only entity

Offering this server to clients or customers would change the answer, so that requires a fresh policy review and a separate design.

Ad content must also follow Google's advertising policies. Humans approve all new ad copy at Tier 2, and disapprovals raise an alert. New campaigns also need the required EU political advertising declaration, which the owner confirms in the approval diff.

Data model and technical requirements

The server's own PostgreSQL database is the system of record for proposals and audit history. Google's change history is only a cross-check, because it covers 30 days and may omit some changes.

Data model

Table	Purpose	Key fields
`config_limits`	Owner-set caps and allowlists; versioned	account ID, caps, domains, geos, autonomy level, version
`goals`	What Claude optimizes for	conversion actions, target CPA or ROAS, protected terms
`agent_runs`	One row per scheduled or chat run	loop, prompt version, start and end, summary
`proposals`	Every requested change	tier, status, diff, reason, expected effect, state hash, expiry
`changes`	Every change sent to Google	proposal, API request ID, old and new values, actor, result
`outcomes`	Effect of each change after its review window	baseline, result, verdict, rollback link
`alerts`	Watchdog findings	type, severity, entity, opened, acknowledged, resolved

Table	Purpose	Key fields
`metric_snapshots`	Daily and intraday metrics for baselines and alerts	date, entity, cost, clicks, conversions, value
`agent_notes`	Short notes carried between runs	topic, text, expiry

The `changes` table is append-only. The application's database role has no update or delete permission on it.

Stack and services

- **Application:** Java 21, Spring Boot, Spring AI MCP server over Streamable HTTP, and the Google Ads Java client.
- **Runtime:** A GKE deployment for the MCP server and approval page, plus a Kubernetes CronJob for the watchdog. Cloud Run is the cheaper alternative if no cluster is already running.
- **Storage:** Cloud SQL for PostgreSQL and Secret Manager.
- **Alerts:** Email plus one push channel, sent directly by the watchdog so alerts work without Claude.
- **Monitoring:** A dead-man's switch that alerts if the watchdog misses two runs, plus metrics for guard rejections, API errors, quota and token refresh failures.

Google Ads API requirements

- Target API v26, released 14 October 2026 with a tentative November 2027 sunset, using Java client 48.0.0 or later.
- Plan four API upgrades a year at most. Google now ships two releases a month, with four majors a year, and only majors can break code.
- Run every mutation with `validate_only` first, then again for real with `partial_failure` off, using the atomic bulk mutate for campaign creation.
- Default new Search campaigns to `target_content_network=false`. Google's own Java sample turns Display Expansion on, which should not be copied by accident.
- Do not send language criteria on Search campaigns; Google rejects them.
- Read budgets with spending limits in mind. A campaign can spend up to twice its average daily budget on a given day, and up to 30.4 times it in a month.

Quota plan

Explorer access covers this design at launch, using about a third of its 2,880 daily operations on production accounts. The operation counts below are planning estimates to confirm during the read-only phase.

Workload	Calculation	Operations per day
Watchdog	96 runs × 5 queries	480
Hourly triage	24 runs × 15 queries	360

Workload	Calculation	Operations per day
Daily and weekly reviews	Estimate	50
Changes, including validation	Estimate	100
Total		**990 (34% of Explorer)**

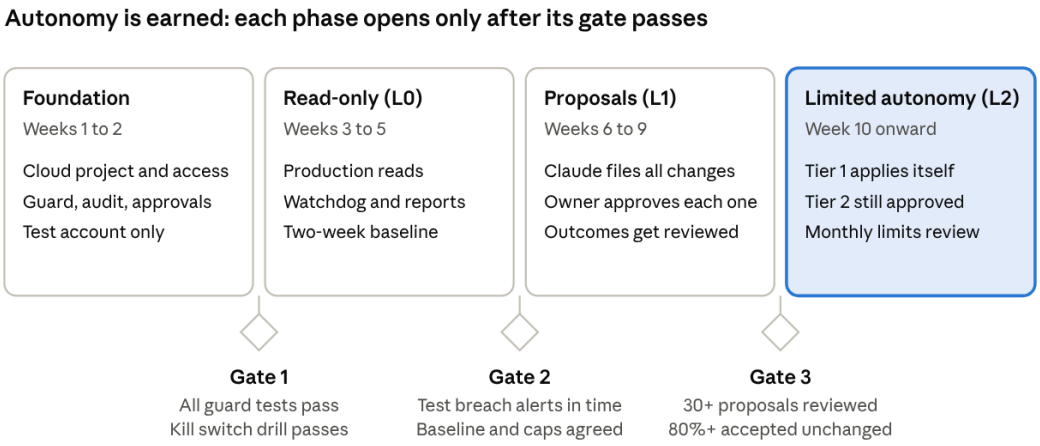
- Requests rejected with a Google Ads failure still count, so refuse bad calls in the guard before they reach Google.
- Quota is measured over a sliding 24 hours. At 80% used, the server pauses Claude's reads and keeps the watchdog running.
- Prefer `SearchStream`, which counts as one operation however many batches it returns.
- Explorer blocks keyword planning, so a keyword ideas tool waits for Basic access and is not in the v1 catalog. Basic allows 15,000 operations per day but requires brand verification.

Quality requirements

- Guard rules have unit and property-based tests, including boundary values for each limit.
- End-to-end tests run against a Google Ads test account, which does not serve ads.
- A monthly drill tests the kill switch, circuit breaker and rollback.
- Claude's prompts and tool descriptions are versioned, and each run records which version it used.

Rollout

Claude earns write access in stages over about 10 weeks of part-time work, and a gate must pass before each stage opens. Durations are estimates, and a phase runs longer if its gate has not passed.



rollout roadmap · four phases and three gates; phases are not to scale

A serious incident moves the system back one level until its cause is fixed and tested.

Before the first line of code

- ☐ Create a dedicated Google Cloud project and enable the Google Ads API.
- ☐ Apply for Explorer access, which is reviewed automatically in most cases.
- ☐ Create a Google Ads test account and a service account with Standard access.
- ☐ Confirm hard caps, goals, allowed domains and protected terms.

Acceptance criteria for v1

- ☐ A breach injected in the test account raises an alert within 30 minutes and trips the breaker.
- ☐ Every limit in the guardrails table has a test proving a call one step over it is refused.
- ☐ Claude has no MCP route to approving a proposal or changing limits.
- ☐ An injected search term telling the agent to raise a budget changes nothing.
- ☐ Replaying a change with the same idempotency key does not change the account twice.
- ☐ Rolling back a change restores the recorded values and adds its own audit record.
- ☐ With the database unavailable, writes are refused and reads continue.
- ☐ The Claude mobile app can read performance and file a campaign proposal, and the owner can approve it from a phone.
- ☐ A scheduled Claude run completes unattended and records its summary.
- ☐ A missed watchdog run alerts the owner.

Risks and open questions

The largest risks are bad optimization on thin data and Google's policy wording, more than any single technical failure.

Risk	Likelihood	Impact	Mitigation
Too little conversion data, so Claude reacts to noise	High for small budgets	High	Minimum data thresholds, excluded recent days, cooldowns and outcome review
Broken conversion tracking makes every decision wrong	Medium	High	Watchdog alerts on sudden conversion drops; changes pause while one is open
Spend reports lag, so the breaker acts after money is spent	Medium	Medium	Set a Google-side budget as the true ceiling; the breaker is a second layer
Prompt injection through search terms or page text	Medium	Medium	Untrusted-data handling, hard limits and no arbitrary network tools
Google's policy is interpreted more strictly	Low	High	Own project and account, no other users, and local stdio mode as a fallback
Claude run fails or scheduled tasks stop	Medium	Low	Watchdog and alerts run independently; missed runs raise an alert

Risk	Likelihood	Impact	Mitigation
API version sunset breaks the server	Low	Medium	Dependency alerts, upgrade routine and contract tests
Service account key leaks	Low	High	Secret Manager, restricted access, rotation and non-admin account role
Approval fatigue leads to rubber-stamping	Medium	Medium	Batch proposals into a daily digest, and show cost and risk first
Claude costs or usage limits exceed the benefit	Medium	Low	Track usage per loop; reduce triage to alert-driven runs

Open questions

- ☐ Which Google Ads account and business goes first, and what is its monthly budget?
- ☐ What is the primary goal: leads, sales or sign-ups? What target CPA or ROAS applies?
- ☐ Is conversion tracking in place and trusted today?
- ☐ What are the hard daily and monthly caps, and in which currency?
- ☐ Which terms, domains and locations go on the protected and allowed lists?
- ☐ Which push channel should carry alerts and approval links?
- ☐ GKE or Cloud Run: is a cluster already running that this can share?
- ☐ Does hourly triage fit the Claude plan's usage limits, or should alerts trigger it through the Agent SDK?
- ☐ Does a direct read of Google's policy text support the secondary reading used here?

Sources

Checked on 3 November 2026. Service limits and access rules can change, so confirm them before building.

- Access levels and permissible use, Google Ads API
- API limits and quotas, Google Ads API
- Get started with Search campaigns, Google Ads API
- Change event, Google Ads API
- Service account workflow, Google Ads API
- Single-user authentication workflow, Google Ads API
- Deprecation and sunset, Google Ads API
- Daily and monthly spending limits, Google Ads Help (search excerpt only)
- Programmatic proxy definition, Google Ads Help, not opened directly
- Google bans programmatic proxies, Relevant Audience
- Two API releases a month from v26, PPC Land (search excerpt only)
- Building custom connectors, Claude Help Center (search excerpt only)
- About connectors, Claude Help Center (search excerpt only)